

# InspectAI ZED

## Low-Cost Edge-AI Visual Defect Detection

MSME Idea Hackathon 6.0

ZERO DEFECT ELECTRONICS MANUFACTURING



STEP 1

### Inspection Setup

A highly affordable, plug-and-play visual capture station designed specifically for MSME workbenches.



#### Low-Cost Cameras

Utilizes standard smartphone cameras or affordable USB webcams instead of expensive industrial optics.



#### Controlled Lighting

Simple LED ring lights eliminate glare and shadows to capture perfectly clear PCB assemblies.

PCB Boards

LED Bulbs

IoT Modules



STEP 2

### Edge-AI Engine

Instant computer vision processing happens locally on a standard laptop or smartphone—no cloud latency.



#### Defect Detection

Algorithms instantly scan for missing components, solder bridges, bent pins, and burn marks.



#### Zero Latency

Local Edge processing guarantees split-second inspection, preventing production line bottlenecks.

Computer Vision

Edge Processing

YOLO / OpenCV



STEP 3

### Smart Interface

A no-code, worker-friendly dashboard that gives clear, unmistakable feedback for quality control.



#### Visual Highlights

The AI draws red bounding boxes directly around the exact location of the defect on the screen.



#### Pass / Fail Decisions

Provides absolute binary decisions (Pass/Fail) to remove human error and fatigue from inspection.

No-Code Setup

Web Dashboard

Custom Templates



STEP 4

### ZED Outcomes

Transforms MSME reputation by ensuring complete compliance with the Zero Defect initiative.



#### Eliminate Rejections

Dramatically cuts down customer returns and rework costs by catching errors before they ship.



#### Digital Traceability

Generates QR-coded digital quality audit reports for every batch to ensure complete accountability.

Zero Defect

Audit Reports

High ROI